Ángel Chandro-Gómez

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I am a final-year PhD researcher at the International Centre for Radio Astronomy Research (ICRAR), The University of Western Australia (UWA), having submitted my thesis and currently awaiting my defence. My research focuses on investigating the mechanisms driving the formation of massive quenched galaxies in the early Universe using cosmological simulations. I have expertise in galaxy formation and evolution, black-hole growth and AGN feedback, connecting simulations with observations, and running and analysing large-volume cosmological simulations. I also have a particular interest in halo-finding and merger-tree-building techniques.

Education

PhD in Astronomy (thesis submitted) Feb 2023 – Dec 2026 (expected)
ICRAR, The University of Western Australia (UWA), Australia.
Thesis: "*Unveiling the formation of early massive quenched galaxies in cosmological simulations*"
Supervisors: Prof. Claudia Lagos and Prof. Chris Power

MSc in Theoretical Physics (Astrophysics and Physics of the Cosmos) Sep 2021 – Jul 2022
Universidad Autónoma de Madrid (UAM), Spain
Thesis: "*Deep learning calibration of galaxy models within a cosmological context*"
Supervisor: Dr. Violeta González Pérez

BSc in Physics Sep 2017 – Jul 2021
Universidad Autónoma de Madrid (UAM), Spain
Dissertation: "*Computational cosmology: galaxy formation in the universe*"
Supervisor: Prof. Alexander Knebe

First-author refereed publications

Square brackets indicate ADS citations as of September 2026

3. "*Unveiling the population of massive quenched galaxies at $z \gtrsim 2$ in the COLIBRE simulations – II. The role of AGN feedback and environment on their emergence*", **Chandro-Gómez** et al., 2026b, MNRAS (10.1093/mnras/stag1632) [1]
2. "*Unveiling the population of massive quenched galaxies at $z \gtrsim 2$ in the COLIBRE simulations – I. Galaxy demographics*", **Chandro-Gómez** et al., 2026a, MNRAS (10.1093/mnras/stag1481) [8]
1. "*On the accuracy of halo merger trees and the consequences on semi-analytic models of galaxy formation and evolution*", **Chandro-Gómez** et al. 2025, MNRAS, 539, 776 [25]

Co-author refereed publications

6. "*Galaxy Phase-Space and Field-Level Cosmology: The Strength of Semi-Analytic Models*", de Santi et al., 2025, ApJ, 1007, 1 [1]
5. "*Connecting Environment, Star Formation History, and Morphology of Massive Quiescent Galaxies at $3 < z < 4$ with JWST*", Kawinwanichakij et al. 2025, ApJ, 997, 1 [25]
4. "*The Galaxy-Active Galactic Nucleus Scaling Relations over 13 Billion Years in SHARK v2.0. I. Supermassive Black Hole Masses*", Bravo et al. 2025, ApJ, 993, 1 [2]
3. "*The diverse star formation histories of early massive, quenched galaxies in modern galaxy formation simulations*", Lagos et al. 2025, MNRAS, 536, 2324 [62]

2. "*Quenching massive galaxies across cosmic time with the semi-analytic model SHARK V2.0*", Lagos et al. 2024, MNRAS, 531, 3551 [79]

1. "*A massive galaxy that formed its stars at* $z \approx 11$", Glazebrook et al. 2024, Nature, 628, 277 [157]

Submitted papers

3. "*Quiescent Galaxies at* $z > 2$ *from CAPERS spectroscopy and their Number densities*", Shen et al., 2026, ApJ (arXiv:2609.20931)

2. "*Bright star-forming galaxies naturally forming at* $z > 10$ *in the Shark semi-analytic model*", Lagos, **Chandro-Gómez** et al., 2026, MNRAS (arXiv:2609.03537)

1. "*First Light And Reionisation Epoch Simulations (FLARES) XX: Comparing semi-analytic models at high-redshift*", Seeyave et al. 2025, MNRAS (arXiv:2511.11982) [2]

Awards

Ken and Julie Michael Prize 2026, ICRAR, UWA (~ 1.250€)	2026
ESO Financial Support – ESO Workshop "*AGN Feedback Across all Scales and Time*" (900€)	2026
IAU Travel Grant – IAU Symposium "*Massive Galaxies across the Universe*" (1.500€)	2025
ASA Student Travel Assistance (~ 850€)	2025
UWA GRS Travel Award (~ 1.100€)	2025
Scholarship for International Research Fees, UWA	2023 – 2026
University Postgraduate Award, UWA	2023 – 2026
MSc Scholarship "*Grant for the promotion of scientific research in MSc UAM 2021*" (4.400€)	2021 – 2022
Scholarship for excellence, Comunidad de Madrid (1.350€, BSc Year 1)	2017 – 2018

Research experience

PhD Researcher Feb 2023 – Dec 2026 (expected)
ICRAR, The University of Western Australia (UWA)
Funded by Scholarship for International Research Fees and University Postgraduate Award.

- Analysed early massive quenched galaxies in the COLIBRE hydrodynamical simulations, comparing theoretical predictions with *JWST* observations and investigating the role of AGN feedback and environment.
- Compared halo-finding and merger-tree algorithms in terms of numerical artefacts and their impact on semi-analytic model outputs.

Research Intern Sep 2021 – Dec 2022
Universidad Autónoma de Madrid (UAM)
Computational Astrophysics group, Theoretical Physics Dept.
Funded by MSc UAM 2021 research grant.

- Developed a deep-learning emulator to explore and calibrate parameters of a semi-analytic model.
- Optimised workflows to run semi-analytic models on the UNIT dark-matter-only simulation.

Summer Research Intern Jul 2021 – Sep 2021
Universidad Autónoma de Madrid (UAM)
Computational Astrophysics group, Theoretical Physics Dept.

- Developed the computational infrastructure to run semi-analytic models on the UAM HPC system.

Research projects & collaborations

- COLIBRE — International collaboration on cosmological hydrodynamical simulations; led two projects on massive quenched galaxies and AGN feedback.

- SHARK — Developer on semi-analytic modelling of galaxy formation and evolution.
- TESSERA — Collaborator contributing semi-analytic model expertise.
- *JWST* OutThere Large Programme — Collaborator on theoretical modelling and simulation–observation comparisons of high-redshift galaxies.
- Co-Investigator on three ongoing *JWST* observing proposals targeting spatially resolved spectroscopy of massive quenched galaxies, spectroscopy of massive quenched galaxies at $z > 5$ and spectroscopy of galaxies at Cosmic Noon. Co-Investigator on a *Roman Space Telescope* observing proposal investigating quenching across environments at Cosmic Noon.
- Co-Investigator on an ongoing supercomputing allocation proposal at the Pawsey Supercomputing Research Centre to model early-Universe galaxy formation and evolution using cosmological simulations.

Conference contributions

ESO workshop "*AGN Feedback Across all Scales and Time*", Munich, Germany (**contributed talk**) 2026
"*The role of AGN feedback in forming early massive quenched galaxies in COLIBRE*"

European Astronomical Society Annual Meeting, Lausanne, Switzerland (**contributed talk**) 2026
"*On the baryon cycle in early massive quenched galaxies in COLIBRE*"

European Astronomical Society Annual Meeting, Lausanne, Switzerland (**poster + short presentation**) 2026
"*How relevant is bursty star formation for early massive quenched galaxies in COLIBRE?*"

Australian National Institute for Theoretical Astrophysics (ANITA) workshop (**contributed talk**) 2026
"*Unveiling the properties and origin of massive quenched galaxies at $z \geq 2$ in the COLIBRE simulations*"

Australian National Institute for Theoretical Astrophysics (ANITA) school (**lecture**) 2026
"*Halo finding and tree building algorithms*"

Australian National Institute for Theoretical Astrophysics (ANITA) school (**teaching assistant**) 2026
"*The semi-analytic model Shark: Hands-on session*"

New York University Abu Dhabi, UAE (**seminar talk**, online) 2025
"*Unveiling the properties and origin of massive quenched galaxies at $z \geq 2$ in the COLIBRE hydrodynamical simulations*"

Cosmic Dawn Center, Denmark (**seminar talk**) 2025
"*Unveiling the properties and origin of massive quenched galaxies at $z \geq 2$ in the COLIBRE hydrodynamical simulations*""

SWIFTcon workshop, Leiden, The Netherlands (**contributed talk**) 2025
"*On the accuracy of dark matter halo merger trees and the consequences for semi-analytic models of galaxy formation*"

Universidad Autónoma de Madrid, Spain (**seminar talk**) 2025
"*Early ($z \geq 2$) massive quenched galaxies: accurate merger tree modelling and insights from COLIBRE*"

European Astronomical Society Annual Meeting, Cork, Ireland (**contributed talk**) 2025
"*The rise of massive-quenched galaxies in the state-of-the-art COLIBRE simulations*"

IAU Symposium "*Massive Galaxies across the Universe*", Naples, Italy (**contributed talk**) 2025
"*The origin and evolution of massive quenched galaxies at $z > 2$ in the state-of-the-art COLIBRE hydrodynamical suite*"

COLIBRE meeting, Leiden, The Netherlands (**contributed talk**, online) 2025
"*Massive quenched galaxies at $z > 2$ in COLIBRE*"

Astronomical Society of Australia Annual Meeting, Perth Hub, Australia (**contributed talk**) 2024

"*On the accuracy of dark matter halo merger trees & consequences for semi-analytical models of galaxy formation*"

Universidad Autónoma de Madrid, Spain (**seminar talk**) 2023
"*Unveiling the formation of massive galaxies across cosmic time*"

Astronomical Society of Australia Annual Meeting, Sydney, Australia (**poster + sparkler**) 2023
"*Light in the dark: quantifying how well we reconstruct the assembly history of dark matter halos*"

Service

Organizer, Computational and Theoretical Research Group Meetings, ICRAR/UWA	2024 – 2026
Volunteer, European Astronomical Society Annual Meeting, Lausanne, Switzerland	2026
Referee, Monthly Notices of the Royal Astronomical Society (MNRAS)	2026
Organizer, ICRAR Student Day (UWA and Curtin University nodes)	2025
Volunteer, Astrofest (astronomy festival)	2023, 2024
Organizer, ICRAR Social Sports Club	2023 – present

Membership

Astronomical Society of Australia	2023 – present

Relevant skills

- Extensive knowledge of running halo-finding and tree-building algorithms (D-Trees, DHalo, HBT-HERONS), semi-analytic models (Shark, Galform); simulation codes (GADGET-2, GADGET-4, SWIFT). Knowledge of Boltzmann solvers (CAMB, CLASS) and initial condition generators (NgenIC, MONOFONIC).
- Experience in processing terabytes of simulation-related outputs using Python (dark-matter-only and hydrodynamical cosmological simulations: FLAMINGO, COLIBRE, P-Millennium, SURFS, UNIT; semi-analytic models: Shark, Galform; halo-finding and tree-building algorithms: D-Trees, DHalo, HBT-HERONS).
- Simulation data visualisation experience with Matplotlib, Swiftsimio, Swiftgalaxy.
- Programming languages: Python, C++, C, Fortran, Matlab.
- High-performance computing (Slurm), parallel computing (OpenMPI, MPI), deep learning (TensorFlow, Keras), version control (Git, GitHub).

References

Prof. Claudia Lagos (PhD supervisor)
International Centre for Radio Astronomy Research (ICRAR), The University of Western Australia (UWA)
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Prof. Chris Power (PhD supervisor)
International Centre for Radio Astronomy Research (ICRAR), The University of Western Australia (UWA)
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